



Year 12 Mathematics (Extension 2)
HSC ASSESSMENT TASK 3
TERM 2, Week 6, 2006

RM

Name: _____

Teacher: _____

Monday 5th June 2006

- Attempt **ALL** questions. Marks may be deducted for careless, insufficient, or illegible work. Calculators may be used. Total possible mark is **50**.
- Begin each question on a new sheet of paper.
- **TIME ALLOWED:** 95 minutes

Question 1:

(18 marks)

- (a) Find the indefinite integral for :-

$$\int \frac{\ln x}{x} dx \quad [1]$$

- (b) By making an appropriate substitution find:-

$$\int x\sqrt{3x-1} dx \quad [2]$$

- (c)

- i) Find real numbers a and b such that for all values of t , [2]

$$\frac{1}{(2-t)(1+2t)} = \frac{a}{2-t} + \frac{b}{1+2t}$$

- ii) Use the substitution $t = \tan\left(\frac{\theta}{2}\right)$ and the identity in part (i) to find [4]

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{3\sin\theta + 4\cos\theta}$$

- (d) Find the indefinite integral for :-

$$\int \frac{x+3}{\sqrt{x^2-2x+5}} dx \quad [4]$$

(Hint : you may need to use the table of standard integrals to complete your answer)

Question 1 continued over page

(e)

- i) Find the indefinite integral for :-

[2]

$$\int \frac{x^3}{x^2 + 1} dx$$

(Hint: degree of numerator is greater than degree of denominator)

- ii) By first integrating using parts and then using the result from part i) above evaluate the definite integral:-

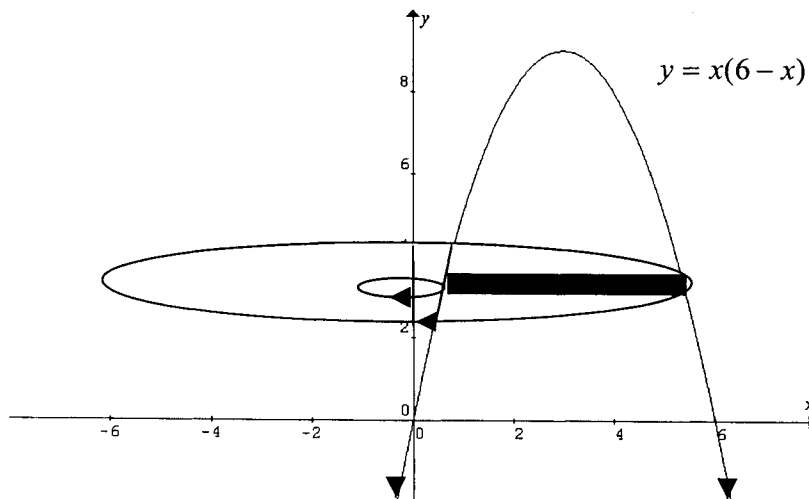
[3]

$$\int_0^1 x^2 \tan^{-1} x dx$$

Question 2: (START NEW PAGE)

(18 Marks)

- (a) The area bounded by the curve $y = x(6 - x)$ and the x -axis is rotated about the y -axis.



- i) Use strips perpendicular to the axis of rotation and show that the x coordinates of the end points of these strips are $3 - \sqrt{9 - y}$ and $3 + \sqrt{9 - y}$.

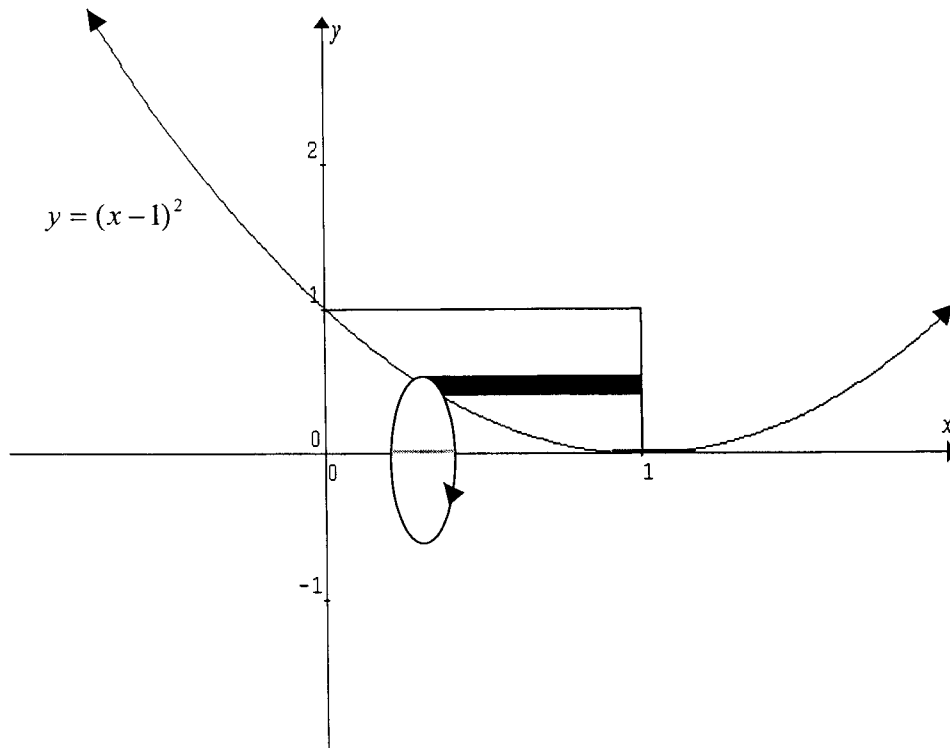
[2]

- ii) Hence find the volume of the solid of revolution in terms of π .

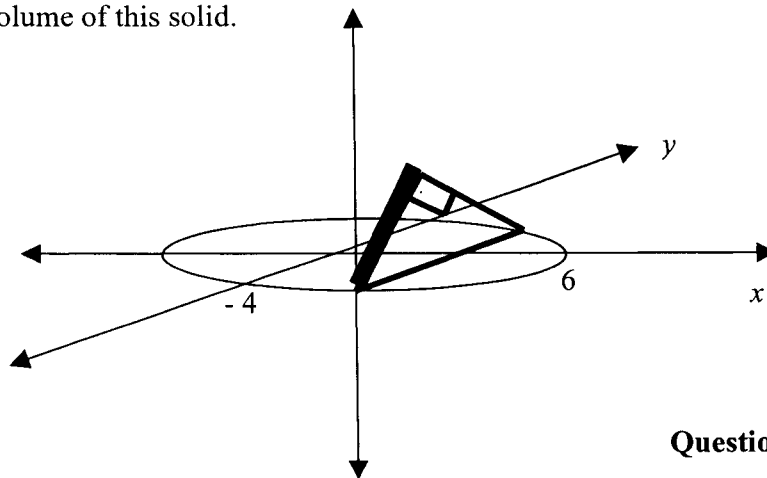
[4]

Question 2 continued over page

- b) The region bounded by the curve $y = (x-1)^2$, $x = 1$, and $y = 1$ is rotated about the x -axis.



- i) Use the method of cylindrical shells to find the volume of one shell δV . [2]
- ii) Hence find the volume (in terms of π) of the solid described above. [4]
- c) The base of a solid is the area of a region bounded by the ellipse whose equation is $4x^2 + 9y^2 = 144$ (note: it cuts the x -axis at 6 and -6 and the y -axis at 4 and -4). Each cross-section of the solid formed by a plane perpendicular to the x and y plane is an isosceles right angled triangle with its hypotenuse in the x and y plane. Find the volume of this solid. [6]

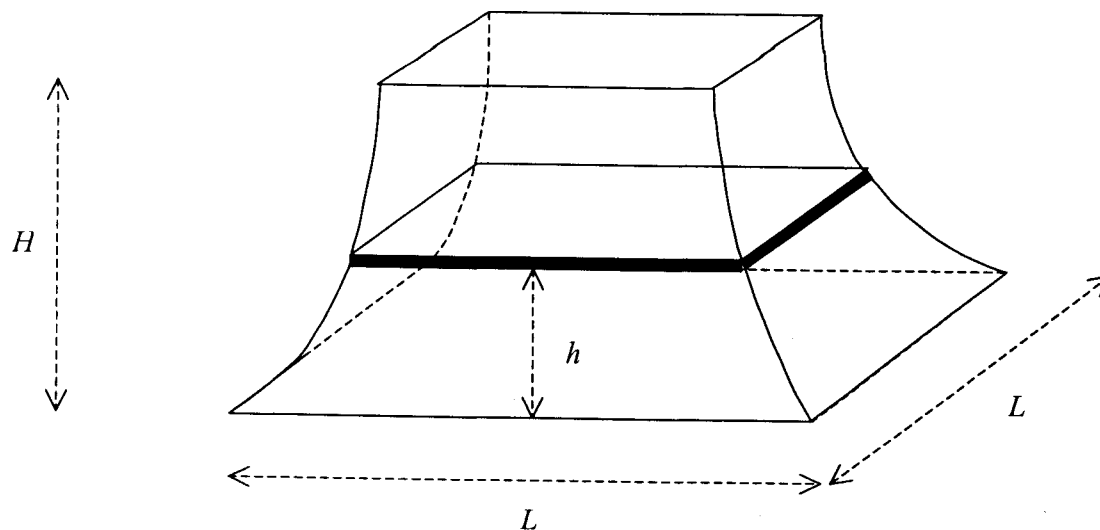


Question 3 over page

Question 3 (START NEW PAGE)

(14 Marks)

a)



A wooden block of height H cm has the shape of a flat-topped square “pyramid” With curved sides as shown above. The cross-section h cm above the base is a square with the sides parallel to the sides of the base and of length $l(h) = \frac{L}{\sqrt{h+1}}$.

Find the volume (correct to the nearest cm) of the wooden block given that $H = L = 30\text{ cm}$. [6]

b)

i) If $I_n = \int \cos^n x \, dx$ show that $I_n = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} I_{n-2}$. [6]

ii) Using the substitution $x = \cos \theta$ and result from part i) above evaluate [4]

$$\int_0^1 \frac{x^3}{\sqrt{1-x^2}} \, dx.$$

END OF TASK

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

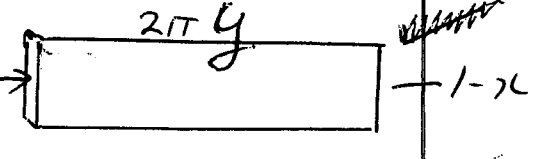
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

Note $\ln x = \log_e x, \quad x > 0$

Year 12 - 2006 Term 2 Mathematics (Ext 2) Assessment Task 3

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
(a) (9) $\int \frac{\ln x}{x} dx$ let $u = \ln x$ $du = \frac{1}{x} dx$ $I = \int u du = \frac{u^2}{2} + C$ $= \frac{(\ln x)^2}{2} + C$ ✓	Can go straight to answer without working	$I = \int_0^1 \frac{1}{3 \left(\frac{2t}{1+t^2} \right) + 4 \left(\frac{1-t^2}{1+t^2} \right)} \times \frac{2 dt}{(t^2+1)}$ $= \int_0^1 \frac{dt}{2+3t-2t^2}$ ✓	
(b) let $u = 3x-1$ $du = 3 dx$ $\therefore I = \frac{1}{3} \int \frac{(u+1)}{3} \sqrt{u} du$ ✓ $= \frac{1}{9} \int u^{3/2} + u^{1/2} du$ $= \frac{2}{9} \left(\frac{(3x-1)^{5/2}}{5} + \frac{(3x-1)^{3/2}}{3} \right) + C$ ✓		$= \frac{1}{5} \int \frac{1}{(2-t)} + \frac{2}{(1+2t)} dt$ $= \frac{1}{5} \left[\ln \left \frac{1+2t}{2-t} \right \right]_0^1$ ✓ $= \frac{1}{5} (\ln 3 - \ln \frac{1}{2})$ $= \frac{1}{5} \ln 6$ ✓	
(c) (i) $1 = a(1+2t) + b(2-t)$ let $t=2 \Rightarrow 1 = 5a \Rightarrow a = \frac{1}{5}$ ✓ $t = -\frac{1}{2} \Rightarrow 1 = 5b \Rightarrow b = \frac{2}{5}$ ✓ ii.) $t = \tan \frac{\theta}{2}$ $dt = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta$ $= \frac{1}{2} (\tan^2 \frac{\theta}{2} + 1) d\theta$ $= \frac{1}{2} (t^2 + 1) d\theta$ $d\theta = \frac{2 dt}{t^2 + 1}$ ✓ $\theta = \frac{\pi}{2} \Rightarrow t = 1$ $\theta = 0 \Rightarrow t = 0$		(d) $I = \int \frac{x+3}{\sqrt{x^2-2x+5}} dx$ $I = \frac{1}{2} \int \frac{2x-2}{\sqrt{x^2-2x+5}} + 4 \int \frac{dx}{\sqrt{x^2-2x+5}}$ ✓ $= \sqrt{x^2-2x+5} + 4 \int \frac{dx}{\sqrt{(x-1)^2+4}}$ ✓ $= \sqrt{x^2-2x+5} + 4 \log (x-1) + \sqrt{x^2-2x+5} + C$ ✓	

Year 12 - 2006 Term 2 Mathematics (Ext 2) Assessment Task 3

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(e) (i) $\int \frac{x^3}{x^2+1} dx$ $= \int x - \frac{x}{x^2+1} dx$ ✓ $= \frac{1}{2}x^2 - \frac{1}{2} \ln(x^2+1) + C$ ✓ (ii) $I = \int_0^1 x^2 \tan^{-1} x dx$ $u = \tan^{-1} x \quad du = \frac{1}{1+x^2}$ $dv = x^2 \quad v = \frac{1}{3}x^3$ $\therefore I = \left[\frac{1}{3}x^3 \tan^{-1} x \right]_0^1 - \frac{1}{3} \int_0^1 \frac{x^3}{1+x^2} dx$ ✓ $= \left[\frac{1}{3}x^3 \tan^{-1} x \right]_0^1 - \frac{1}{3} \int_0^1 x - \frac{x}{x^2+1} dx$ $= \left[\frac{1}{3}x^3 \tan^{-1} x - \frac{1}{6}x^2 + \frac{1}{6} \ln(x^2+1) \right]_0^1$ ✓ $= \frac{1}{12} (\pi - 2 + 2 \ln 2)$ ✓		(OA) $x^2 - 6x - y = 0$ ✓ $x = \frac{6 \pm \sqrt{36-4y}}{2}$ ✓ $= 3 \pm \sqrt{9-y}$ ii) $\delta V = \pi (R^2 - r^2) \delta y$ $= \pi (x_2^2 - x_1^2) \delta y$ $= \pi [(3+\sqrt{9-y})^2 - (3-\sqrt{9-y})^2] \delta y$ ✓ $V = \pi \int_0^9 [(3+\sqrt{9-y})^2 - (3-\sqrt{9-y})^2] dy$ ✓ $= \pi \int_0^9 9 + (9-y) + 6\sqrt{9-y} - 9 + 6\sqrt{9-y} - (9-y) dy$ ✓ $= \pi \int_0^9 12\sqrt{9-y} dy$ ✓ $= 12\pi \left[-\frac{2(9-y)^{3/2}}{3} \right]_0^9$ $= 216\pi$ ✓	
(Q2) (i) $y = 6x - x^2$ (a) $= -(x^2 - 6x + 9) + 9$ ✓ $= -(x-3)^2 + 9$ $\therefore (x-3)^2 = 9-y$ ✓ $x = 3 \pm \sqrt{9-y}$		(b) i.)  $2\pi y$ δy $1-x$ $\delta V = 2\pi y(1-x) \delta y$ ✓	

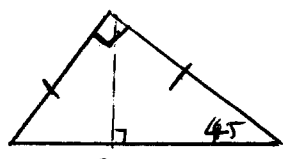
NOTE: $y = (x-1)^2$

$$(x-1) = \pm \sqrt{y}$$

$$x = 1 \pm \sqrt{y}$$

$$\Rightarrow x = 1 - \sqrt{y}$$

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
ii.) $V = 2\pi \int_0^1 y(1-x) dy$ $= 2\pi \int_0^1 y(1-(1-\sqrt{y})) dy$ $= 2\pi \int_0^1 y\sqrt{y} dy$ $= 2\pi \int_0^1 y^{3/2} dy$ $= 2\pi \left[\frac{2y^{5/2}}{5} \right]_0^1$ $= \frac{4\pi}{5} \text{ units}^3$		$= \frac{8}{9} \left[36x - \frac{x^3}{3} \right]_0^6$ $= \frac{8}{9} \times 216 \left[1 - \frac{1}{3} \right]$ $= \frac{8}{9} \times 216 \times \frac{2}{3}$ $= 1280$	
(c)  perp. height = y. $\therefore \text{Area} = \frac{1}{2} (2y)y = y^2$ $\delta V = y^2 \delta x$ $\therefore V = 2 \int_0^6 y^2 dx$ $= \frac{8}{9} \int_0^6 36 - x^2 dx$		Q3 (a) Area of cross-section $= S^2 = \left(\frac{L}{\sqrt{h+1}} \right)^2$ $= \frac{L^2}{h+1}$ $\therefore \delta V = \frac{L^2}{h+1} \delta h$ $V = \int_0^H \frac{L^2}{h+1} dh$ $= L^2 [\ln(h+1)]_0^H$ $= L^2 \ln(H+1)$ $L = H = 30$ $\therefore V = 30^2 \ln 31$ $= 3091 \text{ cm}^3$	

NOTE: $9y^2 = 144 - 4x^2$

$$y^2 = \frac{1}{9} (144 - 4x^2) = \frac{4}{9} (36 - x^2)$$

Year 12 - 2006 Term 2 Mathematics (Ext 2) Assessment Task 3

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Q3 b.) i.) $I_n = \int \cos^n x \, dx$ $= \int \cos^{n-1} x \cos x \, dx$ ✓ $u = \cos^{n-1} x \quad du = (n-1) \cos^{n-2} x \sin x \, dx$ ✓ $dv = \cos x \quad v = \sin x$ $I_n = \cos^{n-1} x \sin x + \int (n-1) \cos^{n-2} x \sin^2 x \, dx$ ✓ $= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx - (n-1) \int \cos^n x \, dx$ ✓ $= \cos^{n-1} x \sin x + (n-1) I_{n-2} - (n-1) I_n$ ✓ $\therefore I_n = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} I_{n-2}$ ✓			
(ii) $x = \cos \theta \quad dx = -\sin \theta \, d\theta$ $x=1 \Rightarrow \theta = 0$ $x=0 \Rightarrow \theta = \frac{\pi}{2}$ $\therefore \int_0^{\frac{\pi}{2}} \frac{x^3}{\sqrt{1-x^2}} \, dx = - \int_{\frac{\pi}{2}}^0 \frac{\cos^3 \theta}{\sqrt{1-\cos^2 \theta}} \times \sin \theta \, d\theta$ ✓ $= \int_0^{\frac{\pi}{2}} \cos^3 \theta \, d\theta = I_3$ $I_3 = \frac{1}{3} \cos^2 x \sin x + \frac{2}{3} I_1$ ✓ $I_1 = \int \cos x \, dx = \sin x$ ✓		$\therefore I_3 = \int \left[\frac{1}{3} \cos^2 x \sin x + \frac{2}{3} \sin x \right]_0^{\frac{\pi}{2}}$ $= \frac{2}{3}$ ✓	